Mikkel Paltorp Schmitt

Scientific Computing · Mathematical Modelling · Programming · Machine Learning · Teaching

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Summary

I'm a self-declared optimist and numbers wizard with a master's degree in mathematical modelling and computing with an emphasis on optimization methods and numerical algorithms from The Technical University of Denmark. During my time at university I have gained a solid background within various applied mathematical topics such as (but not limited to): scientific computing, mathematical optimization and machine learning.

In my spare time I like to go bouldering or sport climbing with my friends. When I am not climbing I volunteer as a route setter at my local climbing gym as I enjoy building problems for other climbers to solve. Other stuff that I also enjoy includes floorball, various eSports, formula 1 and more.

Education

Technical University of Denmark *Sep 2017 - Jan 2020*

M.Sc.Eng in Mathematical Modelling and Computation (GPA: 10.4/12)

Specialization within numerical analysis and optimization methods. Took special and PhD- courses in scientific computing and optimization.

I did my thesis on rank-structured matrices and Gaussian processes (Advisor: Martin S. Andersen).

Technical University of Denmark *Sep 2014 - Sep 2017*

B.Sc.Eng in Mathematics and Technology (GPA: 11.3/12)

Specialization within non-linear dynamics and chaos. Took all M.Sc courses in dynamical systems at DTU Compute.

I did my thesis on regularization of 2D piecewise smooth systems with intersecting discontinuity sets (Advisor: Kristian Uldall Kristiansen).

McGill University *Sep 2016 - Dec 2016*

Exchange semester

Took courses in Mathematical Biology, Non-linear Dynamics, Complex Analysis and Honours Partial Differential Equations.

Work Experience

Technical University of Denmark *Sep 2020 - Present*

PhD Student - Supervisors: Vicente Cutanda Henriquez & Niels Aage

Project: *Efficient Numerical Optimization for Acoustics with Viscous and Thermal Losses and its Application to Acoustic Micro-Devices*

Technical University of Denmark *Sep 2018 - Jan 2020*

Assistant Lecturer (Course 01005)

- Responsibilities included correcting problem sets and supervision/examination of the first year mathematics project.
- Covered topics included complex numbers, linear algebra, (systems of) ODEs, multivariate calculus and simple mathematical modelling.

Technical University of Denmark *Sep 2015 - Jun 2017*

Teaching Assistant (Courses 01901, 01902, 01035, 02601 (ran the Python-to-MatLab tutorial))

- Covered topics included infinite series, stability of non-linear DEs, Fourier series, Fourier transform and different types of convergence.

Projects

Special Course: Methods for Structural Optimization *Collaboration with TICRA*

Together with Martin H. Gæde (Now at TICRA) - Advisors Martin S. Andersen (DTU) & Oscar Borries (TICRA).

- Implemented *the method of moving asymptotes* for minimax optimization in `MatLab` and benchmarked it against TICRAs current solver.

SymEGRSSMatrices.jl

A package for efficiently computing with symmetric extended generator representable semiseparable matrices

- The aim of this project was to derive and implement efficient algorithms for computing with symmetric EGRSS matrices as well as variants.
- All implemented algorithms runs linear in time and covers certain determinants, traces and Cholesky factorizations. At the same time all algorithms only requires the *generators* of the matrices, hence the memory usage is linear, resulting in very scalable computations.
- In this project I learned how to structure a codebase and how to use simple version control (`github`). At the same time I also gained some experience writing automatic tests while using the continuous integration tool `Travis CI`.

Technical Skills

- Proficient: `Julia` (Including the modelling languages `JuMP` and `Convex.jl`), `MatLab`, `Python` and macOS.
- Experience: `C`, `C++`, `R`, `CVX`, `SeDuMi`, `SQL`, `Git` and `Travis CI`.